

# Meeting Minutes - Week 5

**Team name:** TechNova

**Meeting No:** 005

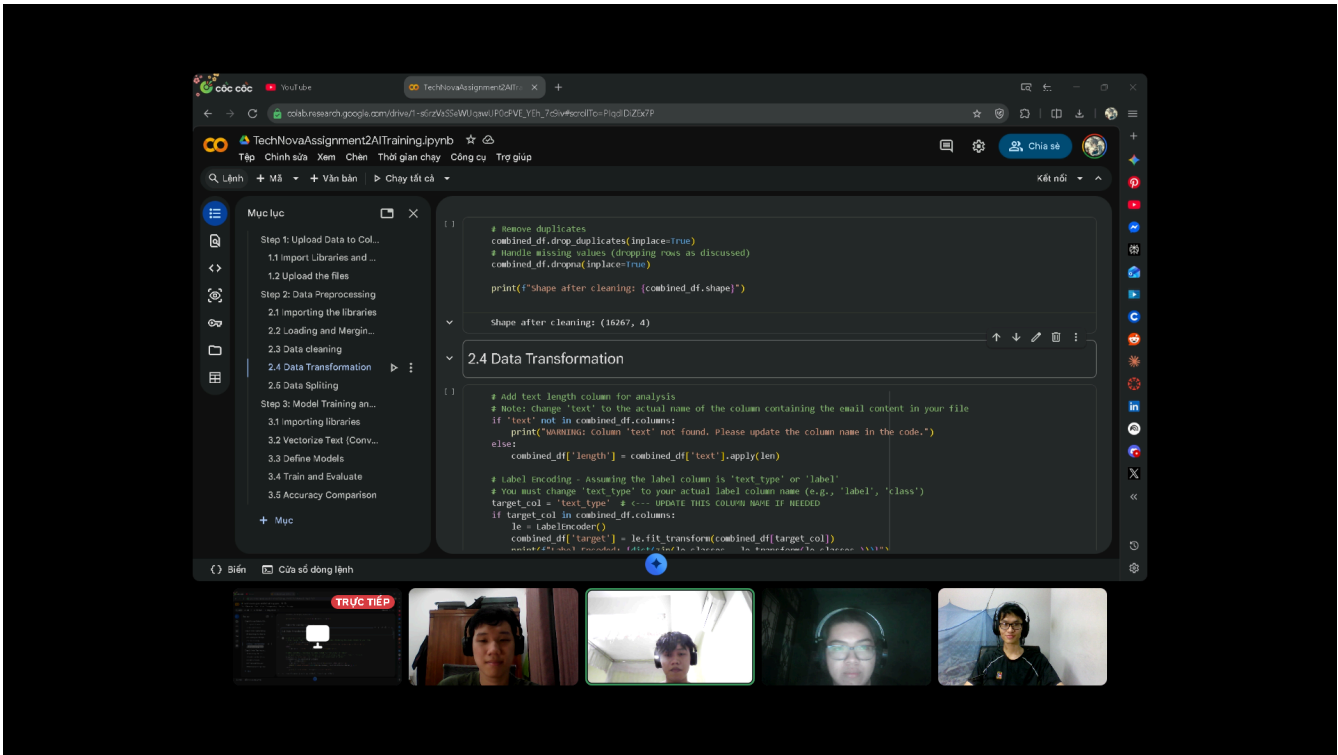
**Location:** Private Meeting in Discord App

**Attending:** Thanh Kien (Leader), Tien Hung, Nam Hung, Huu Hieu

**Apologies:** None. All members are available.

**Meeting start time:** 20/02/2026 - 8:00 p.m. (20:00) UTC +7

**Meeting closed at:** 20/02/2026 - 10:00 p.m. (22:00) UTC +7



**Matters arising from Previous minutes:**

- ☐ Yes
- ☒ No

| Issues raised from previous minutes       | Discussions                                                           | Outcome (Resolved?) |
|-------------------------------------------|-----------------------------------------------------------------------|---------------------|
| Finalizing project scope for Assignment 2 | Clarified that the focus is exclusively on the Spam Detection system. | Resolved            |

|                                               |                                                                                                 |          |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------|----------|
| Setting up the Python development environment | Confirmed all team members have installed the necessary libraries (Pandas, Scikit-learn, etc.). | Resolved |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------|----------|

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**Confirmation of minutes from last meeting:**

- ☒ Yes  
☐ No

**Outcome of meeting:**

| Issue                            | Discussion in brief                                                                                   | Outcome                                                                            | Action: Name and Timeline |
|----------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------|
| ETL Pipeline Implementation      | Discussed the three stages: Extract (raw data), Transform (cleaning), and Load (storing).             | Team aligned on using a structured pipeline for data consistency.                  | February 18 - 20          |
| Data Cleaning Strategy           | Addressed how to handle duplicates, missing values (imputation/masking), and outliers in the dataset. | Agreed to use Pandas for identifying and removing duplicates immediately.          | February 18 - 21          |
| Visualization & Feature Analysis | Evaluated Matplotlib vs. Seaborn for creating heatmaps and pair plots to find correlations.           | Selected both for higher-level statistical graphics and easier outlier detection.  | February 18 - 21          |
| Dataset Preparation              | Discussed splitting data for training and testing to ensure model reliability.                        | Decided to implement the <code>train_test_split</code> function from Scikit-learn. | February 19 - 21          |

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**Actions in brief:**

- The team will prioritize data preprocessing this week.
- Tasks involve cleaning the raw message data, handling missing entries via imputation, and generating correlation heatmaps to visualize feature relationships before moving into model training.

# Meeting Minutes - Week 6

**Team name:** TechNova

**Meeting No:** 006

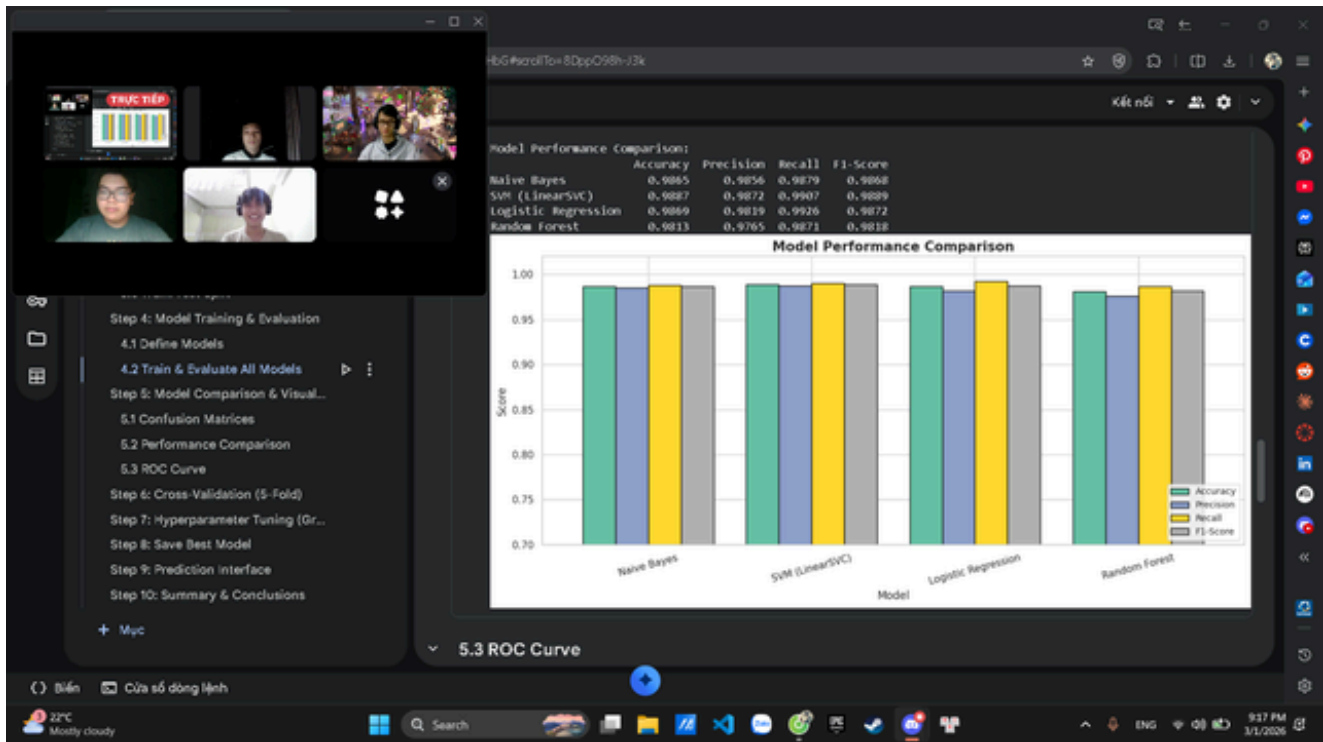
**Location:** Private Meeting in Discord App

**Attending:** Thanh Kien (Leader), Tien Hung, Nam Hung, Huu Hieu

**Apologies:** None. All members are available.

**Meeting start time:** 01/03/2026 - 8:00 p.m. (20:00) UTC +7

**Meeting closed at:** 01/03/2026 - 10:00 p.m. (22:00) UTC +7



**Matters arising from Previous minutes:**

- ☐ Yes
- ☒ No

| Issues raised from previous minutes | Discussions | Outcome (Resolved?) |
|-------------------------------------|-------------|---------------------|
|-------------------------------------|-------------|---------------------|

|                             |                                                                                           |          |
|-----------------------------|-------------------------------------------------------------------------------------------|----------|
| ETL Pipeline Implementation | Discussed the three stages: Extract (raw data), Transform (cleaning), and Load (storing). | Resolved |
| Dataset Preparation         | Discussed splitting data for training and testing to ensure model reliability.            | Resolved |

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**Confirmation of minutes from last meeting:**

- ☒ Yes  
☐ No

**Outcome of meeting:**

| Issue                         | Discussion in brief                                                                                                                | Outcome                                                                            | Action: Name and Timeline |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------|
| Regression Algorithms         | Discussed Simple and Multiple Linear Regression to predict continuous values. Reviewed how features impact the dependent variable. | Selected Linear Regression as a potential model for project performance metrics.   | Feb 24 - March 1          |
| Model Evaluation (Regression) | Reviewed performance metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared ( $R^2$ ).                   | Agreed to use R-squared to evaluate the goodness-of-fit for our regression models. | Feb 24 - March 1          |
| Clustering Methods            | Explored K-Means clustering for grouping data points. Discussed selecting the optimal 'K' using the Elbow Method.                  | Decision made to use clustering for segmenting data patterns within the dataset.   | Feb 25 - March 1          |
| Evaluation of K-Means         | Analyzed Silhouette Score and the Davies-Bouldin                                                                                   | Agreed to implement the Silhouette Score for validating cluster quality.           | Feb 26 - March 1          |

|                                     |                                                                                                     |                                                                                 |                  |
|-------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------|
|                                     | Index to measure cluster cohesion and separation.                                                   |                                                                                 |                  |
| Technical Implementation in Sklearn | Reviewed the code structure for LinearRegression() and KMeans() modules within the Sklearn library. | Team set up the skeleton code for model training based on the lecture examples. | Feb 26 - March 1 |

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**Actions in brief:**

- The team will transition from data preparation to model building.
- Primary focus is on implementing Linear Regression for prediction tasks and K-Means for data segmentation, followed by a rigorous evaluation using MSE and Silhouette scores.

# Meeting Minutes - Week 7

**Team name:** TechNova

**Meeting No:** 007

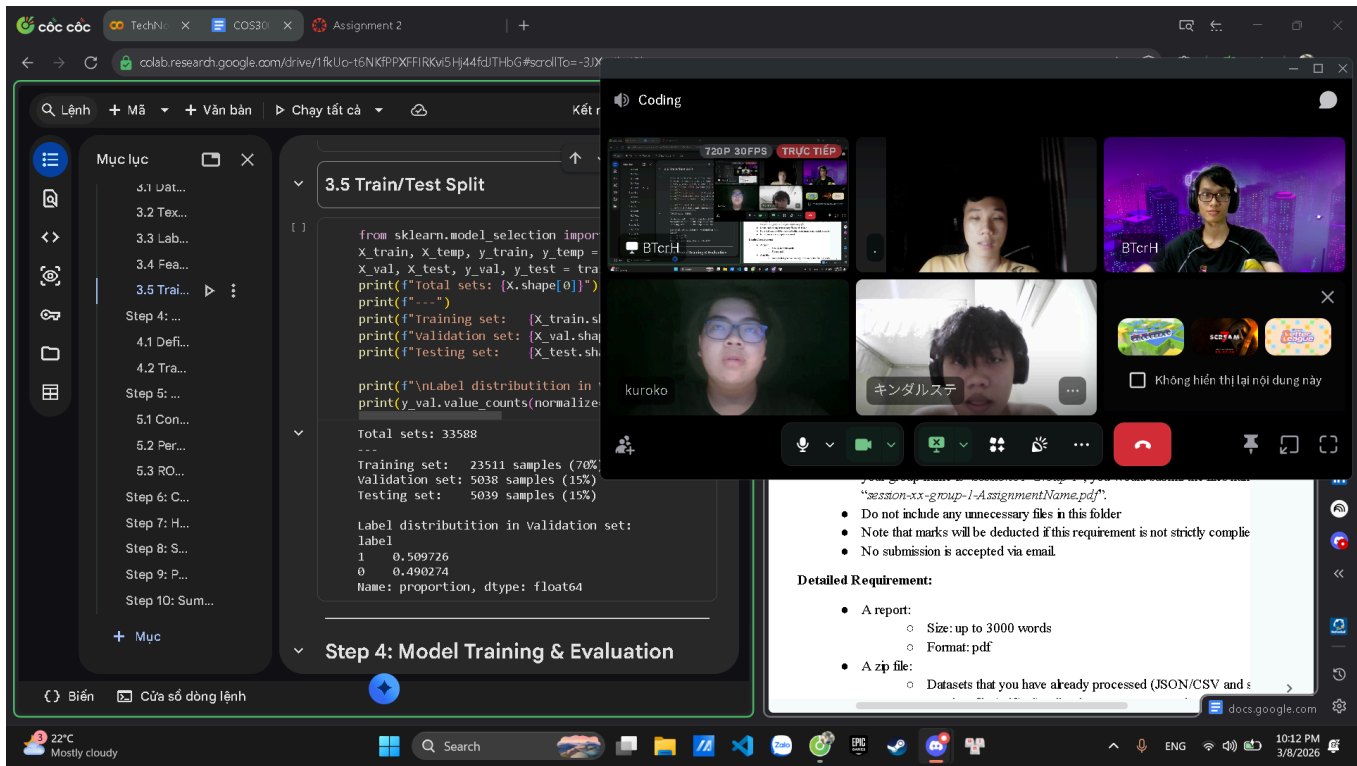
**Location:** Private Meeting in Discord App

**Attending:** Thanh Kien (Leader), Tien Hung, Nam Hung, Huu Hieu

**Apologies:** None. All members are available.

**Meeting start time:** 08/03/2026 - 8:30 p.m. (20:00) UTC +7

**Meeting closed at:** 08/03/2026 - 10:30 p.m. (22:00) UTC +7



**Matters arising from Previous minutes:**

- ☐ Yes
- ☒ No

| Issues raised from previous minutes | Discussions | Outcome (Resolved?) |
|-------------------------------------|-------------|---------------------|
|-------------------------------------|-------------|---------------------|

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|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------|
| Baseline Regression and Clustering Results | Reviewed the outcomes of the Linear Regression and K-Means models developed last week.                        | Resolved. |
| Model evaluation criteria                  | Confirmed that MSE and Silhouette scores are sufficient for the initial regression and clustering benchmarks. | Resolved. |

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**Confirmation of minutes from last meeting:**

- ☒ Yes  
☐ No

**Outcome of meeting:**

| Issue                           | Discussion in brief                                                                                                                              | Outcome                                                                   | Action: Name and Timeline |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------|
| DBSCAN Clustering               | Analyzed the advantages of density-based clustering over K-Means, specifically its ability to handle noise and find arbitrarily shaped clusters. | Agreed to test DBSCAN for identifying anomalies and noise in the dataset. | March 2 - 8               |
| Logistic Regression & KNN       | Evaluated Logistic Regression for binary classification and K-Nearest Neighbors for distance-based classification.                               | Selected both as primary models to compare for the classification tasks.  | March 3 - 8               |
| Advanced Models (Random Forest) | Discussed how ensemble methods like Random Forest reduce overfitting and handle missing data more effectively than single decision trees.        | Decided to implement Random Forest to improve overall model accuracy.     | March 2 - 8               |

|                          |                                                                                                                           |                                                                                                |             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------|
| Classification Metrics   | Reviewed why accuracy can be misleading in imbalanced datasets. Discussed Precision, Recall, F1-Score, and Jaccard Score. | Team agreed to prioritize the F1-Score to balance precision and recall in performance reports. | March 3 - 8 |
| Technical Implementation | Reviewed the Sklearn syntax for LogisticRegression(), KNeighborsClassifier(), and RandomForestClassifier().               | Initial code structure for the classification pipeline was established.                        | March 4 - 8 |

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**Actions in brief:**

- The primary focus for the upcoming week is transitioning from unsupervised clustering to supervised classification.
- We will implement Logistic Regression, KNN, and Random Forest models, ensuring that performance is measured using robust metrics like the F1-Score and Jaccard Index rather than relying solely on accuracy.

# Meeting Minutes - Week 8

**Team name:** TechNova

**Meeting No:** 008

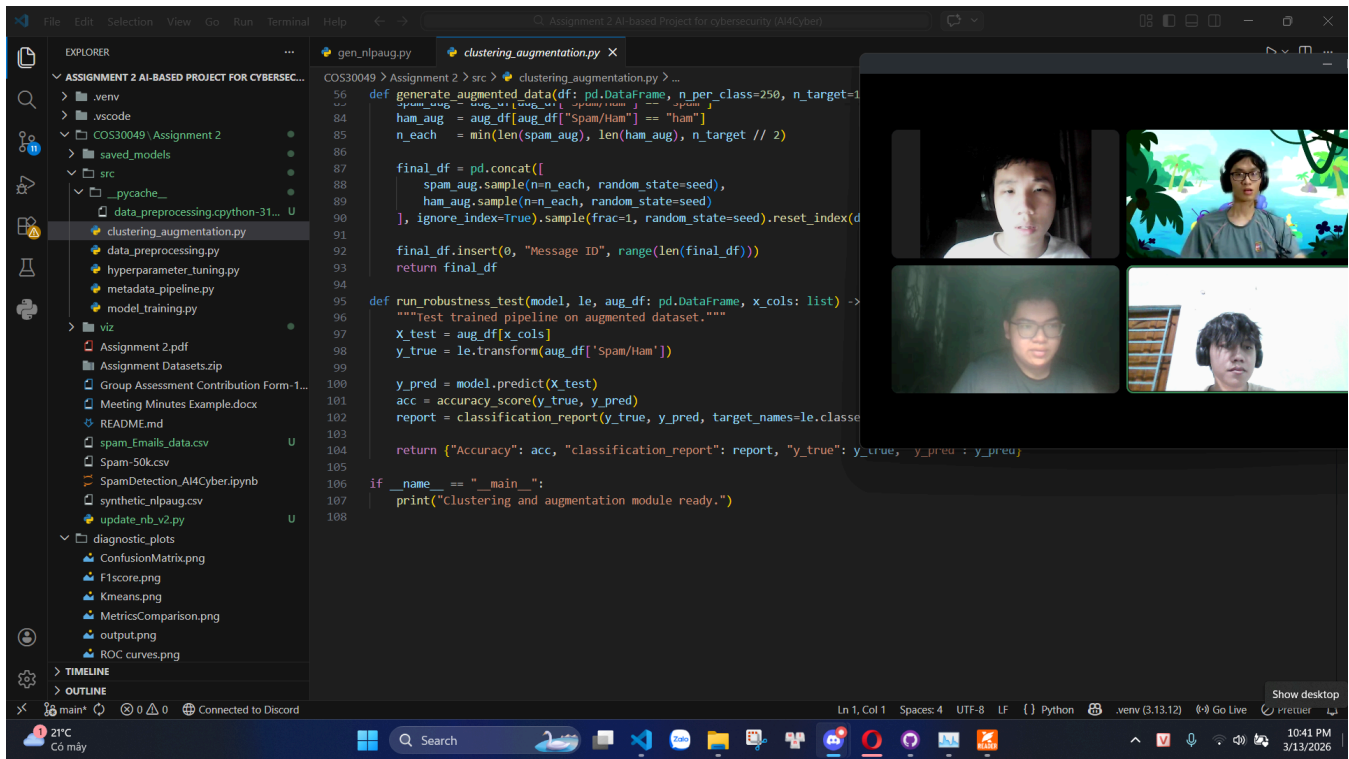
**Location:** Private Meeting in Discord App

**Attending:** Thanh Kien (Leader), Tien Hung, Nam Hung, Huu Hieu

**Apologies:** None. All members are available.

**Meeting start time:** 15/03/2026 - 8:30 p.m. (20:00) UTC +7

**Meeting closed at:** 15/03/2026 - 10:30 p.m. (22:00) UTC +7



**Matters arising from Previous minutes:**

- ☐ Yes
- ☒ No

| Issues raised from previous minutes | Discussions | Outcome (Resolved?) |
|-------------------------------------|-------------|---------------------|
|-------------------------------------|-------------|---------------------|

|                                            |                                                                                      |           |
|--------------------------------------------|--------------------------------------------------------------------------------------|-----------|
| Finalizing Classification model benchmarks | Reviewed the comparison between KNN, Logistic Regression, and Random Forest.         | Resolved. |
| Model performance metrics                  | Confirmed that F1-Score and Jaccard Index have been documented for the final report. | Resolved. |

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**Confirmation of minutes from last meeting:**

- ☒ Yes  
☐ No

**Outcome of meeting:**

| Issue                | Discussion in brief                                                                                                   | Outcome                                                                                     | Action: Name and Timeline |
|----------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------|
| Web App Architecture | Explored the three-layer architecture: Presentation Layer (UI), Application Layer (Logic), and Data Layer (Database). | Agreed on a decoupled architecture to separate the React frontend from the FastAPI backend. | March 9 - 15              |
| Frontend Foundations | Reviewed the roles of HTML (Structure), CSS (Styling), and JavaScript (Functionality) in modern web development.      | Verified that the team has the necessary base knowledge to start building the UI.           | March 10 - 15             |
| React vs. Node.js    | Clarified that React is for frontend UI components while Node.js/npm is the environment used to manage packages.      | Confirmed successful installation of Node.js and initialization of the React environment.   | March 9 - 15              |
| Component-Based UI   | Discussed the benefits of breaking the UI into modular, reusable components for the project's dashboard.              | Identified core components needed: Navigation Bar, Search Input, and Result Display.        | March 10 - 15             |

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**Actions in brief:**

- The team is moving from the machine learning development phase into building the user interface.
- The primary focus for this week is setting up the React framework, designing the modular component structure, and ensuring the presentation layer correctly integrates with the project's logic using HTML and JavaScript foundations.